
Reversing $x + W(x)$

Exact reduction, asymptotic sectors, and a calculus for logarithmic transseries

Central formula

Let W be the principal real Lambert function and let

$$f(x) = x + W(x), \quad x \geq 0.$$

Writing $g = f^{-1}$, $w = W(z)$, and $Lambert_r$ for the inverse of $u \mapsto ue^u + ru$, one has

$$g(z) = z - \mathcal{W}_1(z)$$

and, as $z \rightarrow \infty$,

$$g(z) = z - w + \sum_{k=1}^N \frac{A_k(w)}{z^k} + \mathcal{O}_N\left(\left(\frac{w}{z}\right)^{N+1}\right), \quad A_k(w) = \frac{(-1)^{k+1}}{(k+1)!} \mathcal{L}_{k-1} \cdots \mathcal{L}_0(w^{k+1}),$$

where

$$\mathcal{L}_m = \frac{w}{1+w} \frac{d}{dw} - m.$$

The small parameter is $w/z = e^{-w} \sim (\log z)/z$. This sector is beyond every inverse power of $\log z$.

Abstract

We determine the inverse of $f(x) = x + W(x)$ on the nonnegative real axis and construct its complete asymptotic description at infinity. An elementary change of variable reduces the inverse exactly to the r -Lambert function: if $\mathcal{W}_r$ denotes the inverse of $ue^u + ru$, then $f^{-1}(z) = z - \mathcal{W}_1(z)$. Relative to the ordinary Lambert value $w = W(z)$, the difference is governed by the exponentially small parameter $\eta = w/z = e^{-w}$. We obtain a convergent local expansion in η , a uniform fixed-order remainder estimate, and an explicit differential-operator formula for every coefficient. The first coefficients are rational functions of w , beginning with

$$\frac{w^2}{1+w}, \quad \frac{w^3(w^2-2)}{2(1+w)^3}, \quad \frac{w^4(2w-1)(w^3-6w-6)}{6(1+w)^5}.$$

Combining these sectors with the classical logarithmic expansion of $W(z)$ produces a nested transseries in z^{-1} , $\log z$, and $\log \log z$. We then formulate a general reversal calculus for maps $F(z) = z + \Phi(z)$ whose perturbation is a logarithmic transseries. In the filtered differential algebra with $q = z^{-1}$ and $\delta = z d/dz$, the identity $D(q^m a) = q^{m+1}(\delta - m)a$ turns Lagrange–Bürmann inversion into a triangular, finitely computable algorithm. Fixed-point iteration, Newton iteration, truncation rules, and residual certificates are developed in parallel. The result is both an explicit solution of the stated inverse problem and a reusable method for series reversal across a broad class of polynomial-logarithmic transseries.

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1 Problem, notation, and principal conclusions

Throughout, $W = W_0$ denotes the principal real Lambert function, characterized on $[0, \infty)$ by [1, 3]

$$W(x)e^{W(x)} = x, \quad W(x) \geq 0. \quad (1)$$

We study

$$f(x) = x + W(x), \quad x \geq 0, \quad (2)$$

and its inverse

$$g(z) = f^{-1}(z), \quad z \geq 0.$$

For the asymptotic analysis we use

$$w = W(z), \quad L = \log z, \quad \ell = \log L, \quad q = z^{-1}, \quad \eta = \frac{w}{z} = e^{-w}. \quad (3)$$

The identities $z = we^w$ and $\eta = e^{-w}$ will repeatedly convert ordinary derivatives into operations on rational functions of w .

There are two distinct asymptotic layers.

1. The familiar logarithmic layer is the Poincaré expansion

$$W(z) \sim L - \ell + \frac{\ell}{L} + \frac{\ell(\ell - 2)}{2L^2} + \cdots.$$

Consequently, the entire algebraic-logarithmic expansion of g begins as $z - W(z)$.

2. The difference between $g(z)$ and $z - W(z)$ is not another inverse power of L . It is of order

$$\frac{w^2}{z(1+w)} \sim \frac{\log z}{z},$$

which is smaller than L^{-M} for every fixed M . It therefore belongs to a new, exponentially small transseries sector.

The following statement summarizes the exact and asymptotic answers. The organization by algebra, differentiation, composition, and reversion is also the organizing perspective of the linked ProveIt research-draft directory [8].

Theorem 1.1 (Main inversion theorem). *Let $\mathcal{W}_r$ denote the real inverse, near the nonnegative axis, of*

$$u \mapsto ue^u + ru. \quad (4)$$

Then

$$g(z) = z - \mathcal{W}_1(z), \quad z \geq 0. \quad (5)$$

If $w = W(z)$, then for every fixed integer $N \geq 0$,

$$g(z) = z - w + \sum_{k=1}^N \frac{A_k(w)}{z^k} + O_N\left(\left(\frac{w}{z}\right)^{N+1}\right), \quad z \rightarrow \infty, \quad (6)$$

where

$$A_k(w) = \frac{(-1)^{k+1}}{(k+1)!} \mathcal{L}_{k-1} \mathcal{L}_{k-2} \cdots \mathcal{L}_0(w^{k+1}), \quad \mathcal{L}_m = \frac{w}{1+w} \frac{d}{dw} - m. \quad (7)$$

Moreover,

$$A_k(w) = \frac{w^k}{k} + O(w^{k-1}) \quad (w \rightarrow \infty). \quad (8)$$

Lean status. The exact reduction (5) is formal in `LambertShiftInverse.lean` (`Fabius.lambertShift_lambertShiftInv`, `Fabius.lambertShiftInv_eq_mul_exp`, with $\mathcal{W}_1$ as `Fabius.shiftedLambertW`); the expansion (6)–(8) is not formalized.

The exact reduction in equation (5) is simple, but it does not by itself display the hierarchy of logarithmic and exponentially small terms. The remainder of the article develops that hierarchy and then extracts a general reversal mechanism that no longer depends specifically on W .

2 Exact inversion through the r -Lambert function

2.1 Existence and monotonicity

For $x > 0$, differentiation of equation (1) gives

$$W'(x) = \frac{W(x)}{x(1+W(x))} = \frac{1}{e^{W(x)}(1+W(x))} > 0, \quad (9)$$

with the continuous limiting value $W'(0) = 1$. Hence

$$f'(x) = 1 + W'(x) > 1$$

for $x \geq 0$. The map f is therefore strictly increasing from $[0, \infty)$ onto $[0, \infty)$, so its inverse g exists globally on that interval.

The same monotonicity gives an especially useful a posteriori estimate. If $\tilde{g}(z) \geq 0$ is any approximation and

$$R(z) = f(\tilde{g}(z)) - z, \quad (10)$$

then the mean-value theorem yields

$$|\tilde{g}(z) - g(z)| \leq |R(z)|. \quad (11)$$

Thus a small residual is already a rigorous absolute-error certificate.

2.2 The exact change of variable

Set

$$u = W(g(z)).$$

Then $g(z) = ue^u$, while $z = g(z) + W(g(z))$. Consequently

$$z = ue^u + u. \quad (12)$$

On $u \geq 0$, the right-hand side has derivative $e^u(1+u) + 1 > 0$, so equation (12) has exactly one nonnegative solution. With the notation equation (4), that solution is $u = \mathcal{W}_1(z)$. Since $g = ue^u = z - u$, we obtain

$$g(z) = z - \mathcal{W}_1(z).$$

This proves the exact part of theorem 1.1.

Remark 2.1 (Notation). *The notation follows the generalized Lambert function introduced for the inverse of $ue^u + ru$ in [2]. The subscript in $\mathcal{W}_r$ is a deformation parameter, not a branch index of the ordinary Lambert function. We reserve W for the principal ordinary branch and use calligraphic $\mathcal{W}$ precisely to avoid that ambiguity.*

2.3 A one-parameter family

The same substitution solves a larger family without additional work. For

$$f_r(x) = x + rW(x), \quad r \geq 0, \quad (13)$$

putting $u = W(f_r^{-1}(z))$ gives

$$z = ue^u + ru, \quad f_r^{-1}(z) = ue^u = z - ru.$$

Therefore

$$f_r^{-1}(z) = z - rW_r(z). \quad (14)$$

The original problem is the case $r = 1$. We will retain r in several intermediate formulas because it reveals the organization of the transseries coefficients.

3 The hidden small parameter and an exact difference equation

The ordinary value $w = W(z)$ solves $we^w = z$, whereas $u = W_r(z)$ solves $ue^u + ru = z$. Define

$$d = w - u, \quad \eta = \frac{w}{z} = e^{-w}. \quad (15)$$

Substituting $u = w - d$ into the equation for u , and dividing by $z = we^w$, gives the exact identity

$$\left(1 - \frac{d}{w}\right)(e^{-d} + r\eta) = 1. \quad (16)$$

For fixed large w , this is an implicit analytic equation for d in the small parameter $r\eta$.

The scale η is exponentially small in w , but only algebraically small in z :

$$\eta = e^{-w} = \frac{w}{z} \sim \frac{\log z}{z}. \quad (17)$$

Since $z^{-1} = e^{-L}$, one has

$$\frac{L}{z} = o(L^{-M}) \quad \text{for every fixed } M > 0. \quad (18)$$

This is the precise reason that the correction cannot be detected by carrying the ordinary expansion of $W(z)$ to any finite number of inverse logarithmic orders.

3.1 Analytic expansion in η

Write $\xi = r\eta$ and define

$$H(d, \xi; w) = \left(1 - \frac{d}{w}\right)(e^{-d} + \xi) - 1.$$

At $(d, \xi) = (0, 0)$,

$$H(0, 0; w) = 0, \quad \frac{\partial H}{\partial d}(0, 0; w) = -\left(1 + \frac{1}{w}\right) \neq 0.$$

The analytic implicit-function theorem therefore gives a unique germ

$$d(w, \xi) = \sum_{k \geq 1} b_k(w) \xi^k. \quad (19)$$

For $w \geq 1$, the germ and a sufficiently small disk in ξ may be chosen uniformly: introduce $t = 1/w \in [0, 1]$, observe that $H = (1 - td)(e^{-d} + \xi) - 1$, and apply compactness to the nonvanishing of H_d near $(d, \xi) = (0, 0)$.

The first derivative at the origin is

$$b_1(w) = -\frac{H_\xi}{H_d}(0, 0; w) = \frac{w}{1 + w}. \quad (20)$$

Because

$$f_r^{-1}(z) = z - r(w - d) = z - rw + rd$$

and $\xi = rw/z$, [equation \(19\)](#) becomes

$$f_r^{-1}(z) = z - rw + \sum_{k \geq 1} r^{k+1} \frac{A_k(w)}{z^k}, \quad A_k(w) = w^k b_k(w). \quad (21)$$

For $r = 1$, this is the desired expansion of g .

Theorem 3.1 (Uniform fixed-order remainder). *For each fixed $N \geq 0$ and each bounded interval $0 \leq r \leq R$,*

$$f_r^{-1}(z) = z - rw + \sum_{k=1}^N r^{k+1} \frac{A_k(w)}{z^k} + O_{N,R} \left(\left(\frac{w}{z} \right)^{N+1} \right), \quad z \rightarrow \infty. \quad (22)$$

Proof. The analytic solution $d(w, \xi)$ has a Taylor remainder bounded uniformly for $w \geq 1$ and $|\xi| \leq \rho$, for some fixed $\rho > 0$. Since $\xi = rw/z \rightarrow 0$ uniformly for $0 \leq r \leq R$, truncating [equation \(19\)](#) after N terms gives $d - \sum_{k=1}^N b_k \xi^k = O_{N,R}(\xi^{N+1})$. Multiplication by the outer factor r in $f_r^{-1} = z - rw + rd$ only changes the constant when r is bounded. Substitution of $A_k = w^k b_k$ proves the claim. $\square$

3.2 Leading coefficient at every sector

The difference equation also identifies the dominant term in every z^{-k} sector. Let $w \rightarrow \infty$ in [equation \(16\)](#) while keeping ξ fixed near zero. The limiting equation is

$$e^{-d} + \xi = 1,$$

so

$$d \longrightarrow -\log(1 - \xi) = \sum_{k \geq 1} \frac{\xi^k}{k}. \quad (23)$$

Consequently

$$b_k(w) \longrightarrow \frac{1}{k}, \quad A_k(w) = \frac{w^k}{k} + O(w^{k-1}).$$

The leading pieces of all exponentially small sectors are therefore the Taylor coefficients of a single elementary function, $-\log(1 - \eta)$. This does not replace the full coefficient functions, but it is a useful large- w resummation principle.

4 Lagrange–Bürmann reversal for near-identity maps

The coefficient functions in [equation \(21\)](#) can be obtained directly from a general inversion identity. This section develops the identity first in ordinary differential form and then in the filtered differential algebra adapted to logarithmic transseries.

4.1 The universal derivative formula

Let

$$F(x) = x + \Phi(x) \quad (24)$$

with $\Phi(x) = o(x)$ in the asymptotic regime of interest. Introduce a bookkeeping parameter ε and write

$$G_\varepsilon(z) = z + U_\varepsilon(z), \quad U_\varepsilon = -\varepsilon\Phi(z + U_\varepsilon).$$

Lagrange–Bürmann inversion [4, 5] gives the formal identity

$$G_\varepsilon(z) = z + \sum_{n \geq 1} \frac{(-\varepsilon)^n}{n!} \frac{d^{n-1}}{dz^{n-1}} (\Phi(z)^n). \quad (25)$$

Whenever the right-hand side is summable in the relevant asymptotic filtration, we set $\varepsilon = 1$ and obtain

$$F^{-1}(z) = z + \sum_{n \geq 1} \frac{(-1)^n}{n!} \frac{d^{n-1}}{dz^{n-1}} (\Phi(z)^n). \quad (26)$$

The first four nontrivial blocks are

$$\begin{aligned} F^{-1}(z) = & z - \Phi + \Phi\Phi' - \Phi(\Phi')^2 - \frac{1}{2}\Phi^2\Phi'' \\ & + \Phi(\Phi')^3 + \frac{3}{2}\Phi^2\Phi'\Phi'' + \frac{1}{6}\Phi^3\Phi''' + \dots \end{aligned} \quad (27)$$

For logarithmic Φ , each ordinary derivative introduces one factor of z^{-1} , so the displayed lines are successive transseries sectors.

4.2 The coefficient differential algebra

Set

$$q = z^{-1}, \quad \delta = z \frac{d}{dz}. \quad (28)$$

Let $\mathcal{B}$ be a differential algebra of slowly varying coefficient functions, stable under δ . Typical elements are finite or formal combinations of

$$L = \log z, \quad \ell = \log L, \quad \log \ell, \quad L^{-1}, \quad \ell^{-1}, \dots,$$

with a chosen depth and a chosen asymptotic ordering; this is a deliberately elementary fragment of the differential-algebra viewpoint used for transseries [7]. Work in the filtered ring $\mathcal{B}[[q]]$. For $a \in \mathcal{B}$, direct differentiation gives

$$\frac{d}{dz}(q^m a) = q^{m+1}(\delta - m)a. \quad (29)$$

Thus differentiation raises the q -valuation by at least one. In particular, for $a \in \mathcal{B}$,

$$\frac{d^k a}{dz^k} = q^k (\delta)_{\underline{k}} a, \quad (\delta)_{\underline{k}} = \delta(\delta - 1) \cdots (\delta - k + 1). \quad (30)$$

The operator on the right is a falling factorial in δ .

Theorem 4.1 (Pure logarithmic perturbation). *Suppose $\Phi \in \mathcal{B}$ and $q\Phi \prec 1$. Then the formal inverse of $F(z) = z + \Phi(z)$ is*

$$F^{-1}(z) = z - \Phi(z) + \sum_{k \geq 1} q^k \frac{(-1)^{k+1}}{(k+1)!} (\delta)_k (\Phi^{k+1}). \quad (31)$$

For any prescribed q -order N , only the terms $k \leq N$ can contribute.

Proof. Put $n = k + 1$ in [equation \(26\)](#) and use [equation \(30\)](#). The k -th summand has q -valuation at least k , so coefficient extraction through q^N is finite and triangular. $\square$

Remark 4.2 (General q -dependent perturbations). *If $\Phi = \sum_{j \geq 0} q^j \phi_j$ with $\phi_j \in \mathcal{B}$, the original formula [equation \(26\)](#) still applies. One simply evaluates every derivative by the block rule [equation \(29\)](#) and truncates after each operation. Because the n -th Lagrange term begins at q^{n-1} , terms with $n > N + 1$ cannot affect the inverse through q^N .*

5 Specialization to $\Phi = W$

For $w = W(z)$, [equation \(9\)](#) becomes

$$\delta w = zW'(z) = \frac{w}{1+w}. \quad (32)$$

If $R = R(w)$, then

$$\frac{d}{dz} (z^{-m} R(w)) = z^{-m-1} \left(\frac{w}{1+w} \frac{dR}{dw} - mR \right). \quad (33)$$

This is precisely the operator $\mathcal{L}_m$ in [equation \(7\)](#). Applying [theorem 4.1](#) to $F(x) = x + W(x)$ proves the coefficient formula in [theorem 1.1](#).

5.1 First coefficient functions

The first four rational functions are

$$A_1(w) = \frac{w^2}{1+w}, \quad (34)$$

$$A_2(w) = \frac{w^3(w^2 - 2)}{2(1+w)^3}, \quad (35)$$

$$A_3(w) = \frac{w^4(2w - 1)(w^3 - 6w - 6)}{6(1+w)^5}, \quad (36)$$

$$A_4(w) = \frac{w^5}{24(1+w)^7} (6w^6 - 8w^5 - 69w^4 - 40w^3 + 96w^2 + 72w - 24). \quad (37)$$

The fifth and sixth coefficients, useful for symbolic and numerical checking, are recorded in [section A](#). In general, induction in [equation \(7\)](#) shows that

$$A_k(w) \in \frac{w^{k+1}}{(1+w)^{2k-1}} \mathbb{Q}[w], \quad (38)$$

where the remaining numerator polynomial has degree at most $2k - 2$.

Thus the inverse begins

$$g(z) = z - w + \frac{1}{z} \frac{w^2}{1+w} + \frac{1}{z^2} \frac{w^3(w^2-2)}{2(1+w)^3} + \frac{1}{z^3} \frac{w^4(2w-1)(w^3-6w-6)}{6(1+w)^5} + O\left(\left(\frac{w}{z}\right)^4\right). \quad (39)$$

Because the coefficient functions are evaluated at the exact $w = W(z)$, [equation \(39\)](#) is already a highly effective asymptotic formula without expanding any logarithms.

5.2 Large- w form of the coefficients

Expanding the rational functions at $w = \infty$ gives

$$A_1(w) = w - 1 + \frac{1}{w} - \frac{1}{w^2} + \frac{1}{w^3} + O(w^{-4}), \quad (40)$$

$$A_2(w) = \frac{w^2}{2} - \frac{3w}{2} + 2 - \frac{2}{w} + \frac{3}{2w^2} - \frac{1}{2w^3} + O(w^{-4}), \quad (41)$$

$$A_3(w) = \frac{w^3}{3} - \frac{11w^2}{6} + \frac{23w}{6} - \frac{31}{6} + \frac{31}{6w} - \frac{11}{3w^2} + O(w^{-3}), \quad (42)$$

$$A_4(w) = \frac{w^4}{4} - \frac{25w^3}{12} + \frac{155w^2}{24} - \frac{95w}{8} + \frac{47}{3} + O(w^{-1}). \quad (43)$$

The leading terms agree with [equation \(8\)](#). Retaining only those leading terms in every sector produces

$$g(z) - (z - w) \sim \sum_{k \geq 1} \frac{1}{k} \left(\frac{w}{z}\right)^k = -\log\left(1 - \frac{w}{z}\right),$$

which is the sector-leading resummation already found from the implicit difference equation.

6 The classical logarithmic expansion of W

To express the answer solely through elementary iterated logarithms, write

$$L = \log z, \quad \ell = \log L.$$

Let $\left[\begin{smallmatrix} n \\ j \end{smallmatrix} \right]$ denote the unsigned Stirling number of the first kind. Define

$$P_n(\ell) = \sum_{m=1}^n \frac{(-1)^{n-m}}{m!} \left[\begin{smallmatrix} n \\ n-m+1 \end{smallmatrix} \right] \ell^m. \quad (44)$$

Then the standard asymptotic expansion [\[1, 3\]](#) is

$$W(z) \sim L - \ell + \sum_{n \geq 1} \frac{P_n(\ell)}{L^n}. \quad (45)$$

The first polynomials are

$$\begin{aligned}
P_1(\ell) &= \ell, \\
P_2(\ell) &= \frac{\ell(\ell - 2)}{2}, \\
P_3(\ell) &= \frac{\ell(2\ell^2 - 9\ell + 6)}{6}, \\
P_4(\ell) &= \frac{\ell(3\ell^3 - 22\ell^2 + 36\ell - 12)}{12}, \\
P_5(\ell) &= \frac{\ell(12\ell^4 - 125\ell^3 + 350\ell^2 - 300\ell + 60)}{60}.
\end{aligned} \tag{46}$$

Further polynomials through P_8 are listed in [section B](#).

For each fixed M , the expansion has the usual Poincaré interpretation [6]:

$$W(z) = L - \ell + \sum_{n=1}^M \frac{P_n(\ell)}{L^n} + O\left(\frac{\ell^{M+1}}{L^{M+1}}\right). \tag{47}$$

Since the first beyond-all-orders correction is $O(L/z)$, the purely logarithmic Poincaré expansion of the inverse is

$$g(z) \sim z - L + \ell - \sum_{n \geq 1} \frac{P_n(\ell)}{L^n}. \tag{48}$$

Explicitly,

$$\begin{aligned}
g(z) \sim z - L + \ell - \frac{\ell}{L} + \frac{\ell(2 - \ell)}{2L^2} - \frac{\ell(2\ell^2 - 9\ell + 6)}{6L^3} \\
+ \frac{\ell(12 - 36\ell + 22\ell^2 - 3\ell^3)}{12L^4} - \frac{P_5(\ell)}{L^5} - \dots.
\end{aligned} \tag{49}$$

At every fixed inverse-logarithmic order this is correct, but it cannot reveal the term $A_1(w)/z \sim L/z$. That term is smaller than the remainder of *every* finite truncation in [equation \(49\)](#). This is a standard instance of information that is absent from a single Poincaré scale but present in a transseries.

7 The full nested logarithmic transseries

Combining [equation \(6\)](#) and [equation \(45\)](#) produces the sector decomposition

$$g(z) \sim z + \mathcal{G}_0(L, \ell) + \sum_{k \geq 1} z^{-k} \mathcal{G}_k(L, \ell), \tag{50}$$

where

$$\mathcal{G}_0 = -W_{\log}, \quad \mathcal{G}_k = A_k(W_{\log}), \quad k \geq 1, \tag{51}$$

with $W_{\log}$ denoting the formal series on the right of [equation \(45\)](#).

The ordering must be read sector by sector. All inverse powers of L in $\mathcal{G}_0$ precede the first z^{-1} term; all inverse powers of L in the z^{-1} sector precede the first z^{-2} term; and so on. Symbolically,

$$L^{-M} \gg z^{-1} L^J \quad \text{for every fixed } M, J.$$

This lexicographic ordering is essential. Appending a z^{-1} correction after only a few terms of the outer expansion does not improve a numerical approximation unless the neglected outer remainder has first been made smaller than the new sector.

7.1 First three exponentially small sectors

Straight substitution and re-expansion give

$$\mathcal{G}_1(L, \ell) = L - \ell - 1 + \frac{\ell + 1}{L} + \frac{\ell^2 - 2}{2L^2} + \frac{2\ell^3 - 3\ell^2 - 12\ell + 6}{6L^3} + O\left(\frac{\ell^4}{L^4}\right), \quad (52)$$

$$\mathcal{G}_2(L, \ell) = \frac{L^2}{2} - \frac{2\ell + 3}{2}L + \frac{(\ell + 1)(\ell + 4)}{2} - \frac{(\ell + 1)(\ell + 4)}{2L} + O\left(\frac{\ell^3}{L^2}\right), \quad (53)$$

$$\begin{aligned} \mathcal{G}_3(L, \ell) = & \frac{L^3}{3} - \frac{6\ell + 11}{6}L^2 + \frac{6\ell^2 + 28\ell + 23}{6}L \\ & - \frac{2\ell^3 + 20\ell^2 + 51\ell + 31}{6} + O\left(\frac{\ell^3}{L}\right). \end{aligned} \quad (54)$$

Thus, in fully elementary form,

$$\begin{aligned} g(z) \sim & z - L + \ell - \frac{\ell}{L} + \frac{\ell(2 - \ell)}{2L^2} + \dots \\ & + \frac{1}{z} \left[L - \ell - 1 + \frac{\ell + 1}{L} + \frac{\ell^2 - 2}{2L^2} + \dots \right] \\ & + \frac{1}{z^2} \left[\frac{L^2}{2} - \frac{2\ell + 3}{2}L + \frac{(\ell + 1)(\ell + 4)}{2} + \dots \right] \\ & + \frac{1}{z^3} \left[\frac{L^3}{3} - \frac{6\ell + 11}{6}L^2 + \frac{6\ell^2 + 28\ell + 23}{6}L + \dots \right] + \dots. \end{aligned} \quad (55)$$

The leading term in the k -th small sector is

$$\frac{L^k}{kz^k}. \quad (56)$$

Subleading powers of L and polynomials in ℓ are produced mechanically by [equation \(7\)](#) and [equation \(44\)](#).

7.2 A practical two-level error budget

There are two independent truncation decisions.

- (a) Keeping exact $w = W(z)$ and truncating after the z^{-N} sector has remainder $O((w/z)^{N+1})$ by [theorem 3.1](#).
- (b) Replacing a coefficient $A_k(w)$ by its logarithmic expansion through J subleading inverse powers of L introduces, at that sector, a typical error

$$z^{-k} O(L^{k-J-1} \ell^{J+1}),$$

with the evident modification for the outer sector $k = 0$.

For numerical work, the most efficient representation is usually the exact- W sector expansion [equation \(6\)](#). The fully elementary nested form [equation \(55\)](#) is most useful for structural analysis, comparison of monomials, and integration into a formal transseries system.

8 A general reversal method for logarithmic transseries

We now detach the method from the special function W . Consider

$$F(z) = z + \Phi(z), \quad \Phi \in \mathcal{B}[[q]], \quad q = z^{-1}, \quad (57)$$

where $\mathcal{B}$ is a chosen logarithmic coefficient algebra stable under $\delta = z d/dz$. Assume that $q\Phi \prec 1$, so the map is asymptotically near the identity.

8.1 Direct Lagrange algorithm

The most direct algorithm is a literal implementation of [equation \(26\)](#), with all products and derivatives truncated at the requested q -order.

Algorithm 1: Lagrange–Bürmann reversal through q^N

1. Represent Φ as a truncated block series $\sum_{m=0}^N q^m \phi_m$, with $\phi_m \in \mathcal{B}$.
2. Initialize $G \leftarrow z$.
3. For $n = 1, \dots, N + 1$: form $H \leftarrow \Phi^n$, truncate at q^N , apply $n - 1$ derivatives using

$$D(q^m a) = q^{m+1}(\delta - m)a,$$

truncating after each derivative, and add $(-1)^n H/n!$ to G .

4. Return G truncated through q^N .

Only finitely many coefficient operations are performed, because the n -th summand begins at q^{n-1} .

A compact pseudocode version is as follows.

Listing 1: Filtered Lagrange reversal.

```
def reverse_near_identity(phi, N):
    # phi is a block series sum(q**m * a[m]), truncated at q**N.
    G = z
    for n in range(1, N + 2):
        H = truncate(phi**n, q_order=N)
        for _ in range(n - 1):
            H = truncate(D_block(H), q_order=N)
        G += (-1)**n * H / factorial(n)
    return truncate(G, q_order=N)

# On one block q**m * a:
# D_block(q**m * a) = q**(m+1) * (delta(a) - m*a)
```

8.2 Triangular fixed-point recursion

Writing the inverse as $G = z + U$, one may instead solve

$$U = -\Phi(z + U). \quad (58)$$

Taylor composition gives

$$\Phi(z + U) = \sum_{j \geq 0} \frac{U^j}{j!} \Phi^{(j)}(z). \quad (59)$$

Each derivative raises q -valuation, so the coefficient of q^m on the right depends only on already known lower blocks and the current unknown block linearly. This produces a triangular coefficient recursion. It is often the best method when the coefficient algebra already has a robust composition engine.

8.3 Newton iteration in the filtered algebra

A third method starts from an approximation G_0 , for example $G_0 = z - \Phi(z)$, and iterates

$$G_{j+1} = G_j - \frac{F(G_j) - z}{F'(G_j)}. \quad (60)$$

Since $F' = 1 + o(1)$ is a unit, Newton correction is well-defined in the filtered ring. Under the usual completeness assumptions it approximately doubles the number of correct q -blocks at each step. Newton iteration is therefore preferable at high truncation order, while the Lagrange formula is preferable for closed coefficient formulas and proofs.

8.4 Certification and two-sided checks

For a symbolic candidate G , compute the residual

$$\mathcal{R} = F \circ G - z. \quad (61)$$

If $\mathcal{R} = O(q^{N+1})$, then G is a right inverse through the requested order. A two-sided check also computes $G \circ F - z$. In a well-ordered near-identity class, uniqueness of the inverse makes one side sufficient in principle, but checking both sides is a valuable implementation test for composition, differentiation, and truncation conventions.

For real numerical evaluation, a derivative lower bound converts the residual into an error bound. If $F'(x) \geq m > 0$ on the interval joining $G(z)$ to the exact inverse, then

$$|G(z) - F^{-1}(z)| \leq \frac{|F(G(z)) - z|}{m}. \quad (62)$$

For $F(x) = x + W(x)$, one may take $m = 1$, recovering [equation \(11\)](#).

8.5 Choosing a coefficient algebra and monomial order

A practical transseries implementation must specify more than formulas. It must decide what constitutes a coefficient and how monomials are compared. For the present problem a natural hierarchy is

$$z \gg L \gg \ell \gg 1 \gg L^{-1} \gg \dots \gg z^{-1} L^J \gg z^{-2} L^K \gg \dots \quad (63)$$

for every fixed J, K . More generally, one can use a lexicographic pair (m, ν) , where m is the outer q -sector and ν is the internal logarithmic order. The following rules prevent most implementation errors.

- R1.** Truncate first by outer q -valuation, then by the internal order of coefficient blocks.
- R2.** Keep δ closed on the coefficient algebra; introduce additional iterated logarithms only when differentiation requires them.

- R3.** Apply truncation after every multiplication, derivative, and composition.
- R4.** Record a target remainder ideal, not merely a number of printed terms.
- R5.** Verify the final answer by composition in the same ordering convention.

8.6 A generic logarithmic perturbation

Suppose, for example,

$$\Phi(z) = aL + b\ell + c, \quad L = \log z, \quad \ell = \log L. \quad (64)$$

Then

$$\delta\Phi = a + \frac{b}{L}, \quad \delta^2\Phi = -\frac{b}{L^2},$$

and [equation \(31\)](#) immediately gives

$$\begin{aligned} F^{-1}(z) &= z - \Phi + q\Phi\delta\Phi \\ &\quad - \frac{q^2}{6}(\delta - 1)\delta(\Phi^3) + \frac{q^3}{24}(\delta - 2)(\delta - 1)\delta(\Phi^4) + O(q^4\mathcal{B}). \end{aligned} \quad (65)$$

No ad hoc re-expansion of $\log(z + U)$ or $\log \log(z + U)$ is required. The falling-factorial operators encode all chain-rule and power-of- z bookkeeping automatically. This is the principal advantage of the (q, δ) formulation.

9 Proof of the operator coefficient formula

Although [equation \(7\)](#) follows quickly from Lagrange inversion, it is useful to display the mechanism in full.

For $F(x) = x + W(x)$, [equation \(26\)](#) reads

$$g(z) = z + \sum_{n \geq 1} \frac{(-1)^n}{n!} \frac{d^{n-1}}{dz^{n-1}} w^n. \quad (66)$$

The $n = 1$ term is $-w$. Put $n = k + 1$ in every remaining term. Starting from a function $R(w)$ with no explicit power of z^{-1} , one derivative uses $\mathcal{L}_0$, the next uses $\mathcal{L}_1$, and after k derivatives one has

$$\frac{d^k}{dz^k} R(w) = z^{-k} \mathcal{L}_{k-1} \cdots \mathcal{L}_0 R(w). \quad (67)$$

This follows inductively from [equation \(33\)](#). Taking $R(w) = w^{k+1}$ in [equation \(67\)](#) and substituting into [equation \(66\)](#) gives

$$\frac{(-1)^{k+1}}{(k+1)!} z^{-k} \mathcal{L}_{k-1} \cdots \mathcal{L}_0 (w^{k+1}),$$

which is exactly $z^{-k} A_k(w)$.

The Lagrange series and the analytic η -series are the same expansion in different coordinates. Indeed, $d = w - \mathcal{W}_1(z) = g(z) - (z - w)$, and [equation \(6\)](#) gives

$$d = \sum_{k \geq 1} \frac{A_k(w)}{z^k} = \sum_{k \geq 1} \frac{A_k(w)}{w^k} \eta^k.$$

Comparison with [equation \(19\)](#) shows

$$b_k(w) = \frac{A_k(w)}{w^k}. \quad (68)$$

Thus the implicit-function proof supplies convergence and a remainder bound, while Lagrange inversion supplies the closed differential-operator formula. Together they give more than either method alone.

10 Numerical verification

The exact value can be computed without directly inverting f : solve

$$ue^u + u = z \quad (69)$$

for its nonnegative root and set $g = z - u$. Let

$$G_N(z) = z - w + \sum_{k=1}^N \frac{A_k(w)}{z^k}, \quad w = W(z), \quad (70)$$

with $G_0 = z - w$. High-precision evaluation gives the absolute errors in [table 1](#).

Table 1: Absolute errors $|G_N(z) - g(z)|$ using exact $w = W(z)$.

z	$N = 0$	$N = 1$	$N = 2$	$N = 3$	$N = 4$
10	1.1202×10^{-1}	1.0462×10^{-3}	2.9897×10^{-4}	2.3371×10^{-5}	6.8889×10^{-7}
100	2.6355×10^{-2}	2.1862×10^{-4}	9.4975×10^{-7}	2.3486×10^{-8}	7.9461×10^{-10}
10^3	4.4172×10^{-3}	7.5875×10^{-6}	1.3534×10^{-8}	1.7161×10^{-11}	1.1027×10^{-14}
10^6	1.0464×10^{-5}	4.9551×10^{-11}	2.9302×10^{-16}	1.8408×10^{-21}	1.1635×10^{-26}

Representative exact values are

$$g(10) = 8.366493829844153615806835 \dots,$$

$$g(100) = 96.64072495463040645894733 \dots,$$

$$g(10^3) = 994.7548143481392806156898 \dots,$$

$$g(10^6) = 999988.6166523780211244849 \dots$$

The error pattern is consistent with the scale $(w/z)^{N+1}$. At $z = 10^6$, for example, the zeroth approximation $z - W(z)$ misses a term of order 10^{-5} , while four correction sectors reduce the error below 10^{-26} .

The numerical experiment also illustrates the distinction between the outer and inner expansions. Increasing the number of inverse-logarithmic terms in $W(z)$ drives a log-only approximation toward $z - W(z)$, not toward the exact inverse beyond the floor

$$g(z) - (z - W(z)) = W(z) - \mathcal{W}_1(z) \sim \frac{W(z)^2}{z(1 + W(z))}.$$

Crossing that floor requires the z^{-1} sector.

11 Extensions and variants

11.1 The full r -family

Combining [equation \(14\)](#) with the coefficient construction gives

$$f_r^{-1}(z) = z - rw + \sum_{k \geq 1} r^{k+1} \frac{A_k(w)}{z^k}, \quad w = W(z), \quad (71)$$

with the fixed-order remainder stated in [theorem 3.1](#). Equivalently,

$$\mathcal{W}_r(z) = w - \sum_{k \geq 1} r^k \frac{A_k(w)}{z^k}. \quad (72)$$

Thus one sequence A_k simultaneously describes every bounded value of the parameter r .

11.2 Affine-scaled Lambert perturbations

Consider the three-parameter map

$$F(x) = \alpha x + \beta W(\gamma x), \quad \alpha, \beta, \gamma > 0. \quad (73)$$

Putting $u = W(\gamma x)$ gives

$$\frac{\gamma z}{\alpha} = ue^u + \frac{\beta\gamma}{\alpha}u.$$

Therefore

$$F^{-1}(z) = \frac{z - \beta \mathcal{W}_{\beta\gamma/\alpha}(\gamma z/\alpha)}{\alpha}. \quad (74)$$

The asymptotic expansion follows from [equation \(72\)](#) after replacing z by $\gamma z/\alpha$. This exact reduction is useful when normalizing more complicated inverse problems before invoking the general transseries machinery.

11.3 Perturbations built from iterated logarithms

The differential-algebra method does not require a special-function reduction. For maps such as

$$F(x) = x + a \log x + b \log \log x + \frac{c_0 + c_1 \log \log x}{\log x} + \frac{d_0}{x} + \dots,$$

one places the slowly varying expressions in $\mathcal{B}$, the explicit powers of x^{-1} in the outer q -filtration, and applies either Algorithm 1, the fixed-point recursion, or Newton iteration. The same block derivative [equation \(29\)](#) guarantees triangularity. The only additional work is to define a consistent internal ordering for the chosen depth of iterated logarithms.

12 Implementation checklist and common pitfalls

The following checklist summarizes the method in a form suitable for a computer-algebra implementation or a semi-formal proof development.

A robust reversal workflow

1. **Normalize the map.** Rewrite it as $F = z + \Phi$, possibly after an affine change of variables.
2. **Choose the outer parameter.** Usually $q = z^{-1}$; verify that $q\Phi \prec 1$.
3. **Choose the coefficient algebra.** Fix the allowed logarithmic depth and ensure closure under $\delta = zD_z$.
4. **State the ordering.** Use outer q -valuation first and logarithmic order second.
5. **Generate coefficients.** Use Lagrange–Bürmann for closed formulas, a triangular fixed point for transparent recursion, or Newton for high order.
6. **Truncate after every operation.** This keeps all computations finite and prevents irrelevant high sectors from expanding combinatorially.
7. **Compose back.** Check $F \circ G - z$ to the target remainder ideal; preferably also check $G \circ F - z$.
8. **Certify numerically.** Where a derivative lower bound is available, convert the residual directly into an error estimate.

Several mistakes recur in logarithmic reversal problems.

- P1. Mixing scales.** A finite $1/L$ expansion cannot resolve a term of size L/z . The latter requires a new sector.
- P2. Differentiating coefficients as constants.** The operator $D(q^m a) = q^{m+1}(\delta - m)a$ contains both δa and the derivative of q^m .
- P3. Expanding too early.** Keeping $w = W(z)$ exact makes the sector coefficients compact and numerically stable. Expand in L, ℓ only when an elementary nested form is required.
- P4. Using a one-dimensional truncation counter.** A transseries remainder is an ideal in a multi-scale ordering, not merely “the next printed term.”
- P5. Ignoring branch data.** The present real nonnegative problem has unique branches, but complex or negative-domain variants require explicit branch selection before composition and inversion.

13 Conclusions

The inverse of $x + W(x)$ admits both an exact special-function representation and a transparent complete asymptotic structure. The exact representation is

$$f^{-1}(z) = z - \mathcal{W}_1(z).$$

With $w = W(z)$, the practically useful expansion is

$$f^{-1}(z) = z - w + \frac{w^2}{z(1+w)} + \frac{w^3(w^2 - 2)}{2z^2(1+w)^3} + \cdots,$$

and truncation after z^{-N} has error $O((w/z)^{N+1})$. The parameter $w/z = e^{-w}$ identifies the correction as beyond all orders of the conventional inverse-logarithmic expansion. Re-expansion of each rational coefficient in $L = \log z$ and $\ell = \log L$ yields the full nested logarithmic transseries.

The broader reversal method is encoded by two formulas:

$$F^{-1}(z) = z + \sum_{n \geq 1} \frac{(-1)^n}{n!} D^{n-1}(\Phi^n), \quad F = z + \Phi,$$

and

$$D(q^m a) = q^{m+1}(\delta - m)a, \quad q = z^{-1}, \quad \delta = zD.$$

Together they make coefficient extraction finite, triangular, and compatible with arbitrary logarithmic depth. The same structure supports direct Lagrange inversion, fixed-point recursion, Newton acceleration, and residual certification. The Lambert example is therefore not only solvable in closed form; it is a model case for systematic reversal of polynomial-logarithmic transseries.

A Higher rational coefficient functions

For compactness, write

$$A_5(w) = \frac{w^6}{120(1+w)^9} Q_5(w), \tag{75}$$

where

$$Q_5(w) = 24w^8 - 58w^7 - 428w^6 - 151w^5 + 1320w^4 + 1380w^3 - 480w^2 - 720w + 120.$$

The sixth coefficient is

$$A_6(w) = \frac{w^7}{720(1+w)^{11}} Q_6(w), \quad (76)$$

with

$$Q_6(w) = 120w^{10} - 444w^9 - 2920w^8 + 424w^7 + 16577w^6 + 17892w^5 - 13530w^4 - 26400w^3 - 1800w^2 + 7200w - 720.$$

They are generated without solving any new implicit equations: one repeatedly applies $\mathcal{L}_m$ according to [equation \(7\)](#).

A minimal symbolic implementation is

Listing 2: Generating $A_k(w)$ symbolically.

```
from sympy import symbols, diff, factorial, factor
w = symbols('w')

def L_operator(m, expr):
    return factor(w/(1+w) * diff(expr, w) - m*expr)

def A(k):
    expr = w**(k+1)
    for m in range(k):
        expr = L_operator(m, expr)
    return factor((-1)**(k+1) * expr / factorial(k+1))
```

B Logarithmic coefficient polynomials through order eight

Using [equation \(44\)](#), the next polynomials after those in [equation \(46\)](#) are

$$P_6(\ell) = \frac{\ell}{120} (20\ell^5 - 274\ell^4 + 1125\ell^3 - 1700\ell^2 + 900\ell - 120), \quad (77)$$

$$P_7(\ell) = \frac{\ell}{840} (120\ell^6 - 2058\ell^5 + 11368\ell^4 - 25725\ell^3 + 24500\ell^2 - 8820\ell + 840), \quad (78)$$

$$P_8(\ell) = \frac{\ell}{2520} (315\ell^7 - 6534\ell^6 + 45962\ell^5 - 142149\ell^4 + 205800\ell^3 - 135240\ell^2 + 35280\ell - 2520). \quad (79)$$

Together with [equation \(46\)](#), these determine the outer sector through L^{-8} . Substitution into the rational A_k generates any desired number of internal logarithmic terms in the exponentially small sectors.

C Independent derivation from the implicit difference equation

For completeness, one may compute the same coefficients without Lagrange inversion. Substitute

$$d = \sum_{k=1}^N b_k(w) \xi^k + O(\xi^{N+1})$$

into

$$\left(1 - \frac{d}{w}\right) (e^{-d} + \xi) = 1,$$

expand the exponential, and equate powers of ξ . The coefficient of ξ^k is linear in b_k , because the derivative of the left-hand side with respect to d at the origin is $-(1 + 1/w)$. Thus the recursion is triangular. The first three results are

$$\begin{aligned} b_1(w) &= \frac{w}{1+w}, \\ b_2(w) &= \frac{w(w^2 - 2)}{2(1+w)^3}, \\ b_3(w) &= \frac{w(2w-1)(w^3 - 6w - 6)}{6(1+w)^5}. \end{aligned}$$

Multiplying by w^k recovers A_k through [equation \(68\)](#). This route is especially convenient for proving convergence in η ; the operator route is more convenient for symbolic generation.

D Reference identities for verification

The following identities form a compact test suite for an implementation.

$$\begin{aligned} W(z) e^{W(z)} &= z, \\ \delta W(z) &= \frac{W(z)}{1+W(z)}, \\ \mathcal{W}_r(z) e^{\mathcal{W}_r(z)} + r \mathcal{W}_r(z) &= z, \\ f_r^{-1}(z) &= z - r \mathcal{W}_r(z), \\ \left(1 - \frac{W(z) - \mathcal{W}_r(z)}{W(z)}\right) \left(\exp(\mathcal{W}_r(z) - W(z)) + r \frac{W(z)}{z}\right) &= 1, \\ F(F^{-1}(z)) - z &= 0. \end{aligned}$$

For a truncation through z^{-N} , the final identity should reduce to the target remainder $O((W(z)/z)^{N+1})$, either symbolically or after asymptotic normalization.

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